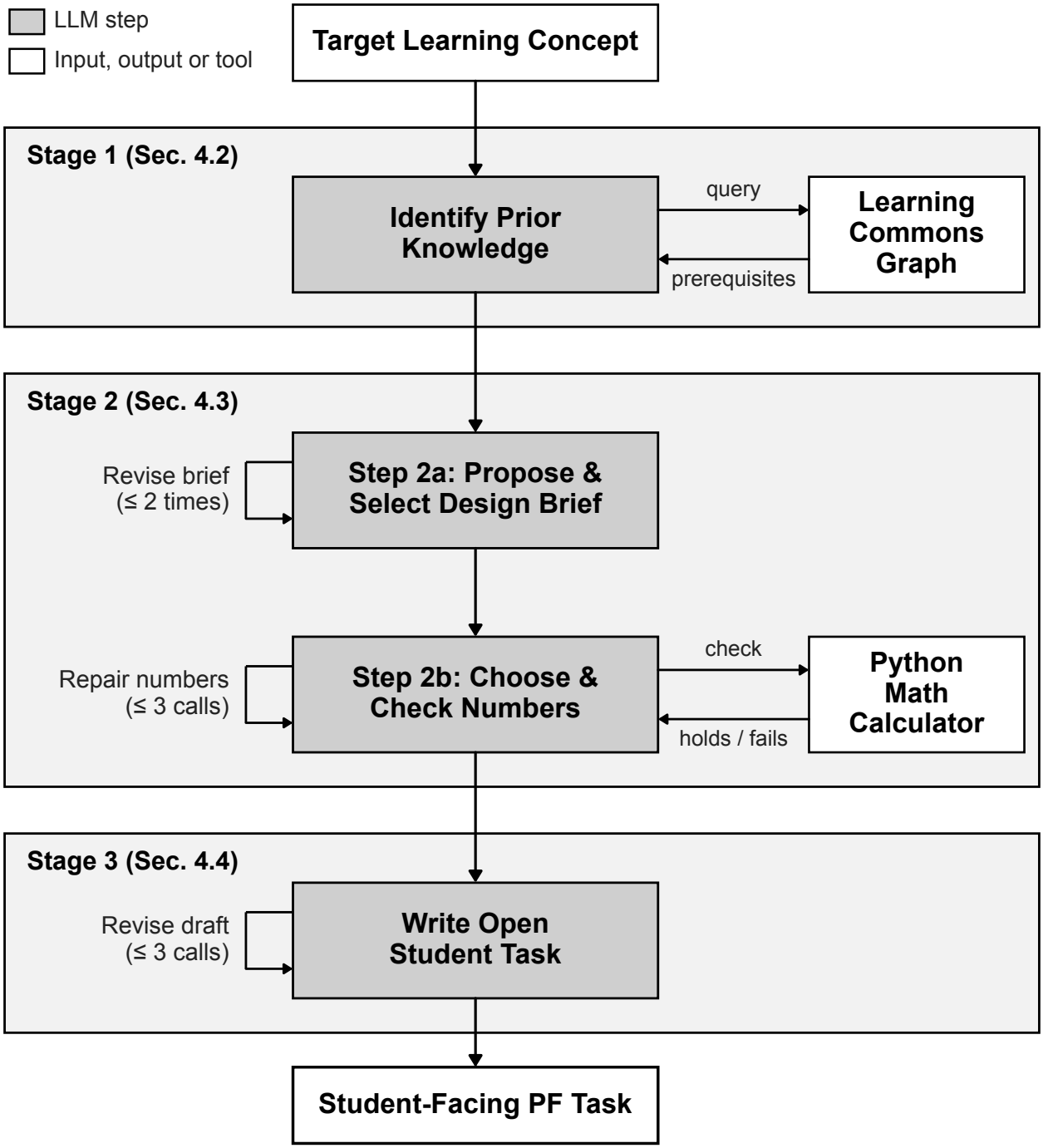
LLM step
Input, output or tool
Target Learning Concept
Stage 1 (Sec. 4.2)
Identify Prior Knowledge
query
prerequisites
Learning Commons Graph
Stage 2 (Sec. 4.3)
Revise brief (≤ 2 times)
Step 2a: Propose & Select Design Brief
Repair numbers (≤ 3 calls)
Step 2b: Choose & Check Numbers
check
holds / fails
Python Math Calculator
Stage 3 (Sec. 4.4)
Revise draft (≤ 3 calls)
Write Open Student Task
Student-Facing PF Task